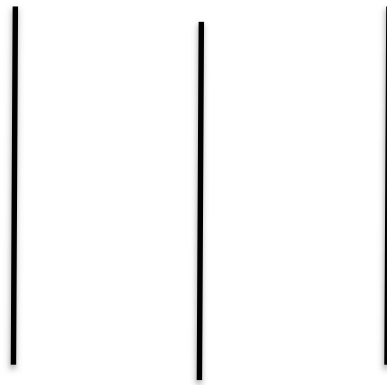



**Universitat
de Girona**


IFROS 2024-2026
Autonomous System
Lab3: Graph Search (A*)



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1. Introduction

1.1. Graph Search

Graph search is a fundamental algorithmic technique used to explore or traverse a graph, which is a collection of nodes (or vertices) connected by edges. The goal of graph search is to find specific nodes, paths, or properties of the graph, such as:

- Determining if a path exists between two nodes.
- Finding the shortest path or least costly route.
- Discovering all reachable nodes from a starting point.
- Identifying cycles, components, or connected structures.

1.2. A* Algorithm

The A* algorithm is a type of informed graph search algorithm used for finding the shortest path between a starting node and a target node. A* uses a cost function to decide which node to explore next.

The cost function is:

$$f(n) = g(n) + h(n)$$

- **$f(n)$** : The total estimated cost of the cheapest solution that passes through node n .
- **$g(n)$** : The actual cost to reach node n from the start node.
- **$h(n)$** : The heuristic estimate of the cost to reach the goal from node n .

The algorithm selects and expands the node with the smallest $f(n)$, balancing two aspects:

- **Exploration**: Captured by $g(n)$, the cost incurred to reach the current node from the start.
- **Goal-Directed Behaviour**: Captured by $h(n)$, the estimated cost to the goal from the current node.

1.3. Connectivity (4 point and 8 point)

In grid-based representations (such as images or maps), connectivity defines how neighboring pixels or cells are connected. These connections influence graph traversal, distance calculations, and pathfinding algorithms.

In **4-point connectivity**, each cell (or pixel) is connected to its **4 direct neighbors**: up, down, left, and right. Diagonal neighbors are not considered connected. The distance between each 4-connected neighbor is usually 1 unit (Manhattan distance).

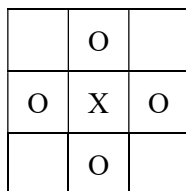


Fig-1: 4-point connectivity

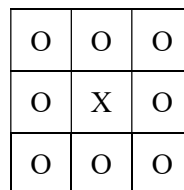


Fig-2: 8-point connectivity

1.4. Lab Activity

In the lab session, first we work on extracting datasets from the csv file for environment and visibility, then we implemented A* algorithm to find shortest path from start to goal position. Similarly, later we implemented A* on Grid map provided to find the shortest path.

2. Implementation

2.1 Datasets on .csv format

Code Organization in .ipynb file

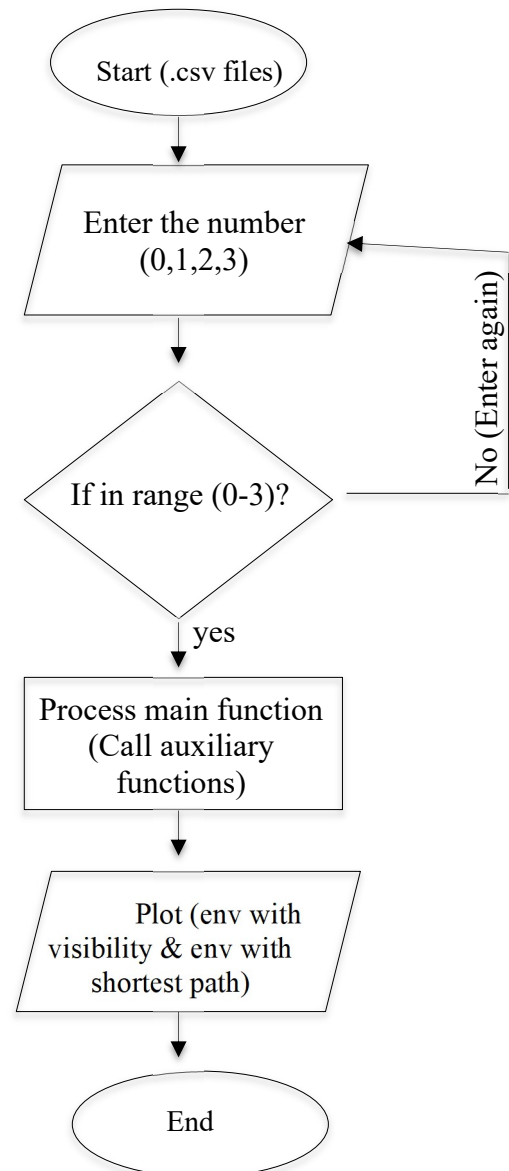
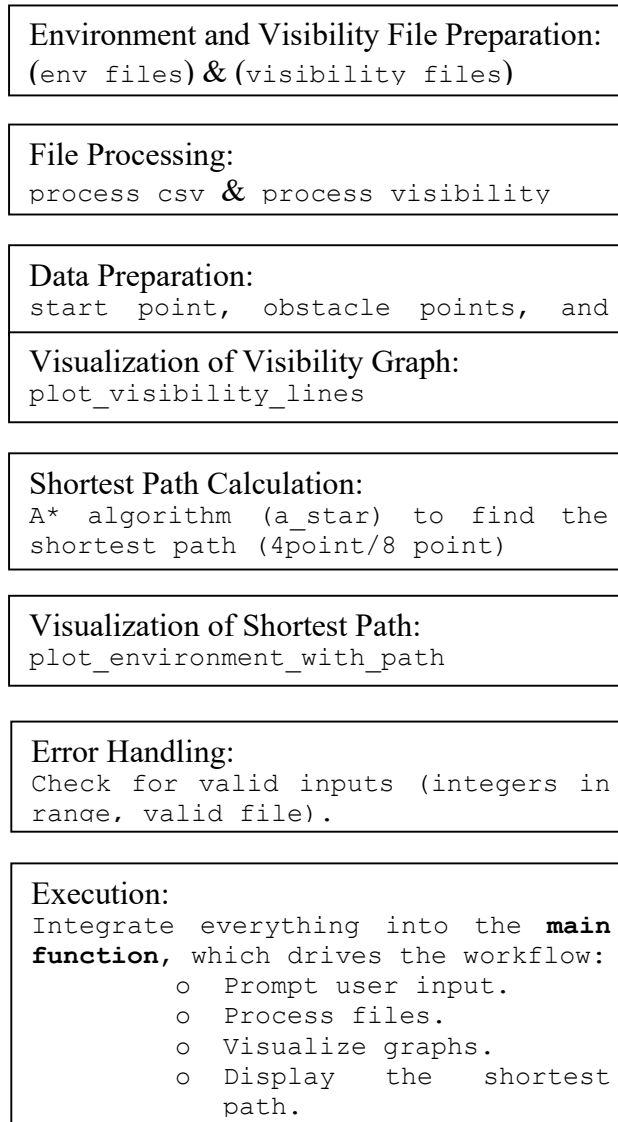


Fig-3: flowchart for csv files

Workflow:

- Input: User selects environment (0–3).
- Parse: Load and process environment and visibility graph data.
- Prepare: Merge all points into a graph structure.
- Visualize: Display visibility connections.
- Compute: Find and display the shortest path with A*.
- Output: Annotated plots of the environment and path, along with shortest path.

2.2 Grid Map

Code Organization in .ipynb file

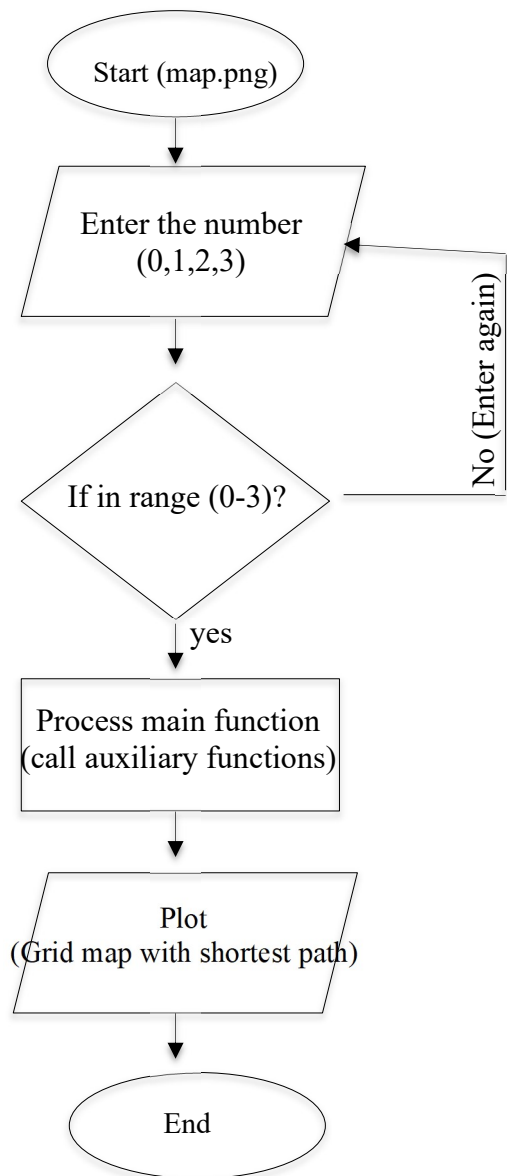
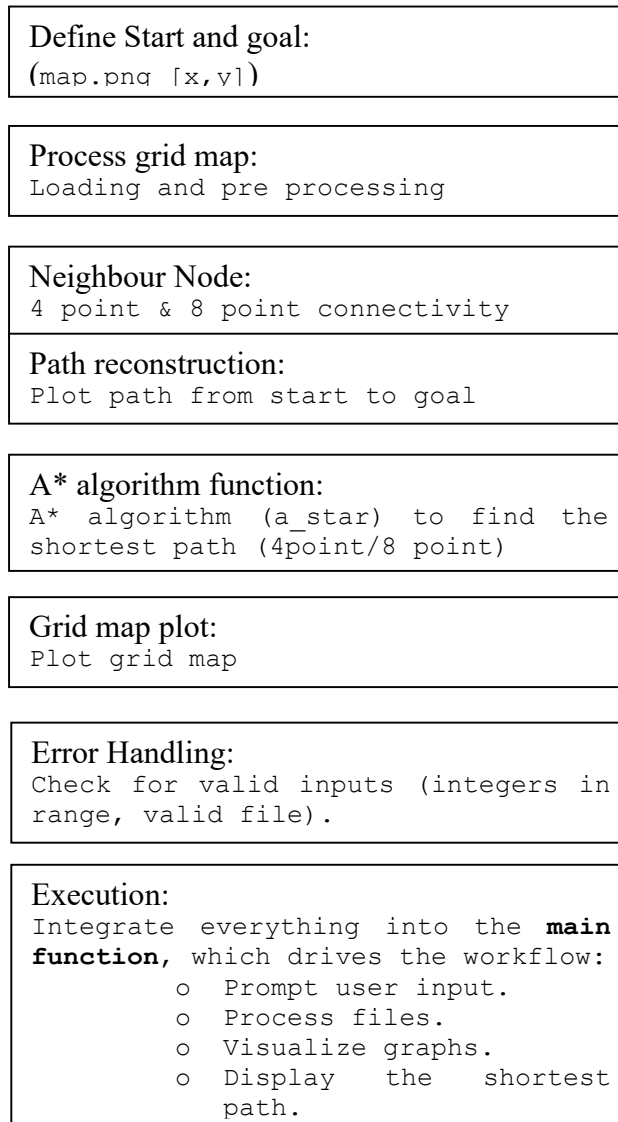


Fig-3: flowchart for grid map

Workflow:

- Input: User selects environment (0–3).
- Parse: Load and process grid map, start and goal.
- Compute: Find and display the shortest path with A*.
- Output: Annotated plots of the grid map, along with shortest path.

3. Output & Result

3.1 .csv files

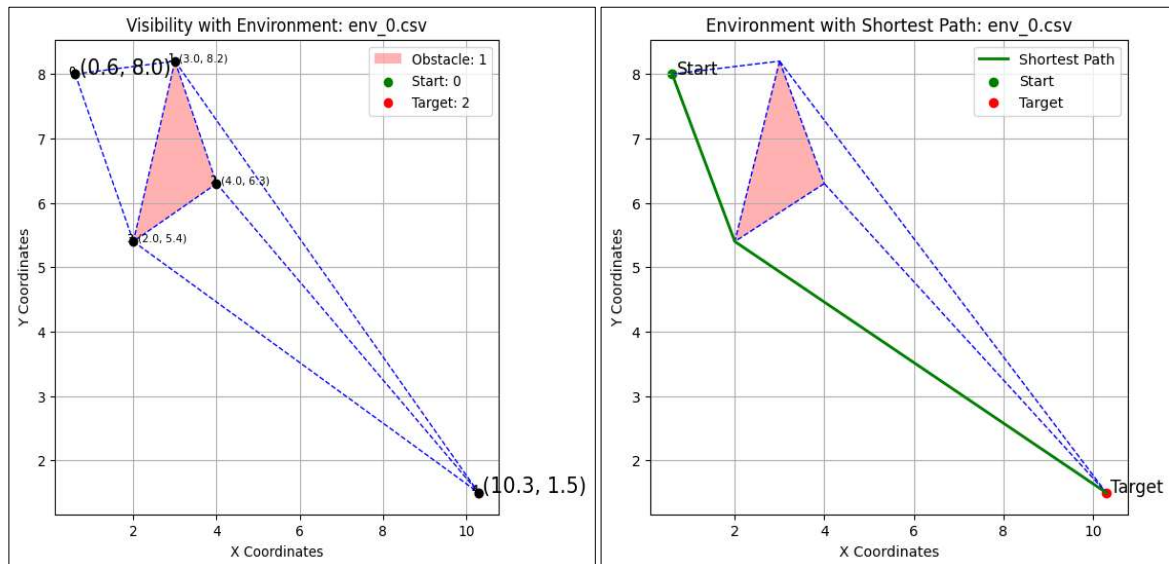
3.1.1 env_0.csv and visibility_graph_env_0.csv

Inter a number between 0 and 3 to select environment and visibility: **0 (Enter)**
 Start Coordinates: (0.6, 8.0)
 Target Coordinates: (10.3, 1.5)
 Obstacles: {'1': [(3.0, 8.2), (4.0, 6.3), (2.0, 5.4)]}

Points: {0: (0.6, 8.0), 1: (3.0, 8.2), 2: (4.0, 6.3), 3: (2.0, 5.4), 4: (10.3, 1.5)}

Visibility: [(0.0, 1.0), (0.0, 3.0), (1.0, 3.0), (1.0, 2.0), (1.0, 4.0), (2.0, 3.0), (2.0, 4.0), (3.0, 4.0)]

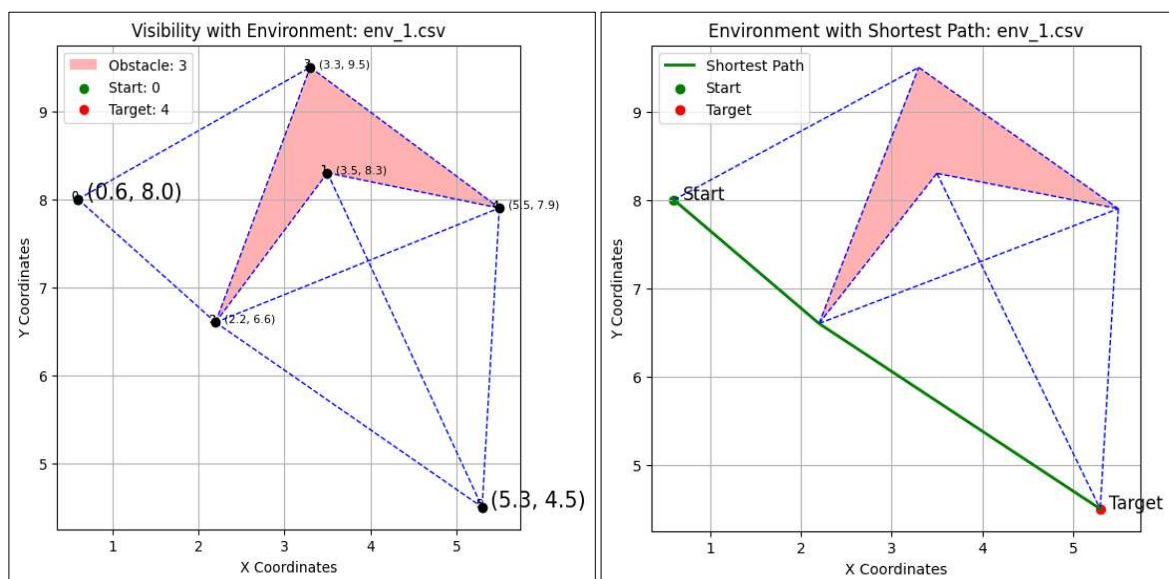
Shortest Path: [0, 3.0, 4.0]



3.1.2 env_1.csv and visibility_graph_env_1.csv

Enter a number between 0 and 3 to select environment and visibility: 1 (Enter)

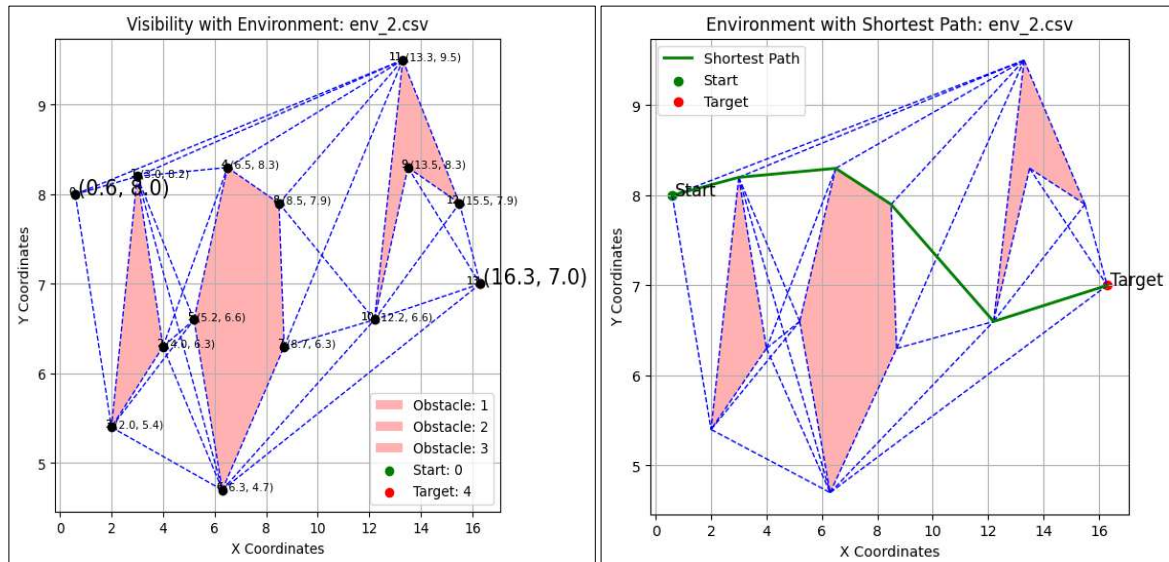
(Please refer to Appendix 3.1.2 env_1.csv and visibility_graph_env_1.csv for output values)



3.1.3 env_2.csv and visibility_graph_env_2.csv

Enter a number between 0 and 3 to select environment and visibility: 2 (Enter)

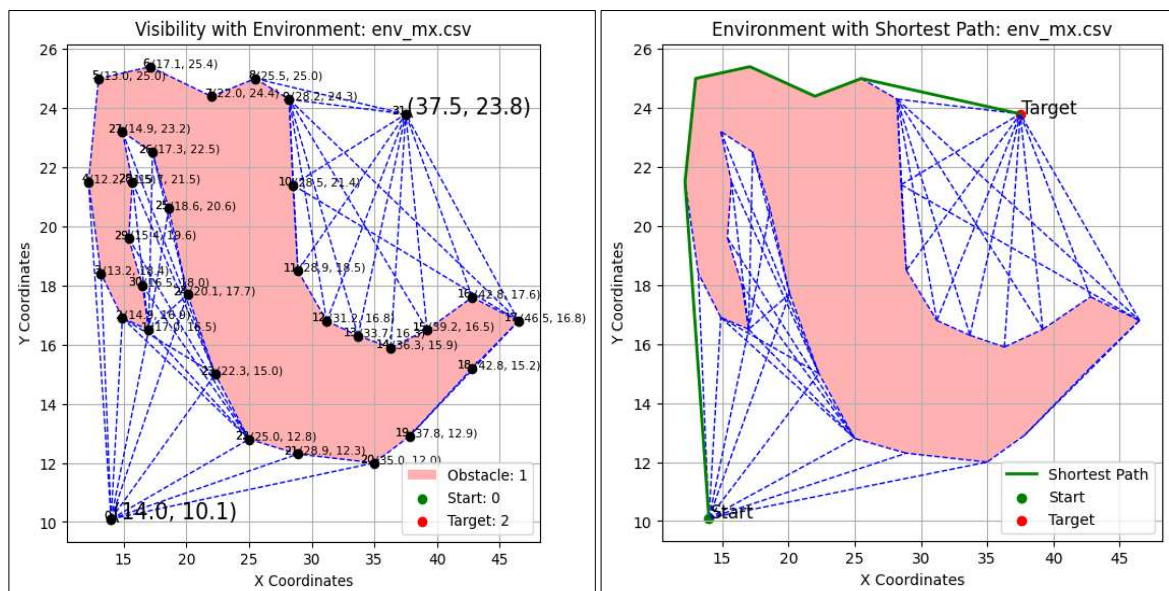
(Please refer to Appendix 3.1.3 env_2.csv and visibility_graph_env_2.csv for output values)



3.1.4 env_mx.csv and visibility_graph_env_mx.csv

Enter a number between 0 and 3 to select environment and visibility: 3 (Enter)

(Please refer to Appendix 3.1.4 env_mx.csv and visibility_graph_env_mx.csv for output values)

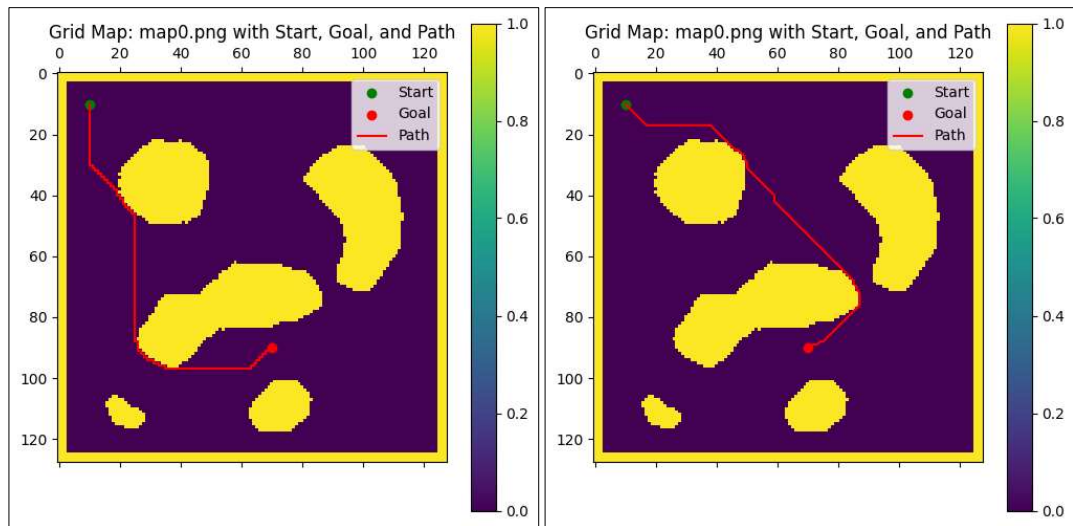


3.2 Grid Map

3.2.1 map_0.png

Inter a number between 0 and 3 to select environment and visibility: **0 (Enter)**

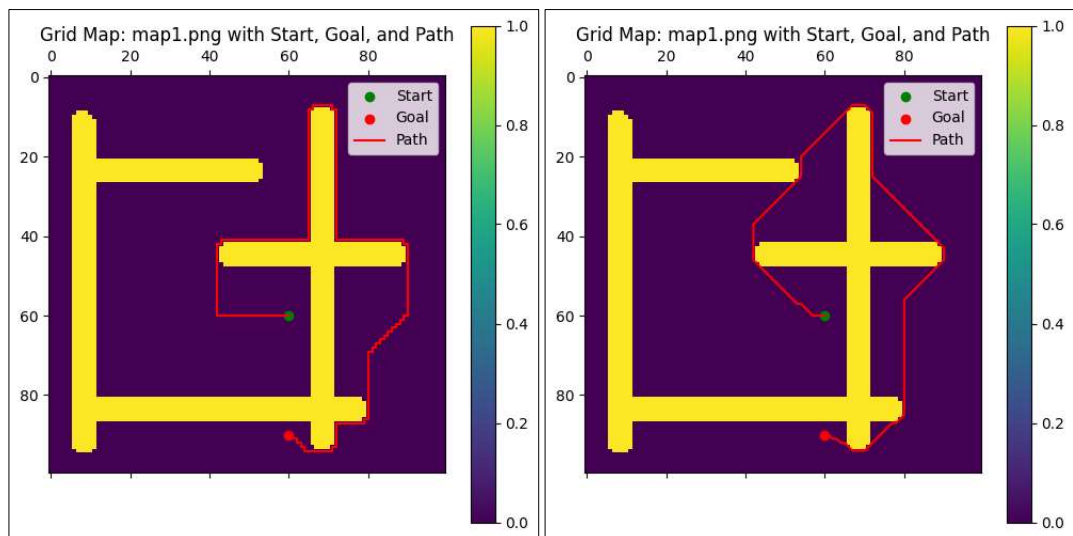
(Please refer to Appendix 3.2.1 map_0.png for output values of 4 point and 8 point connectivity with cost value)



3.2.2 map_1.png

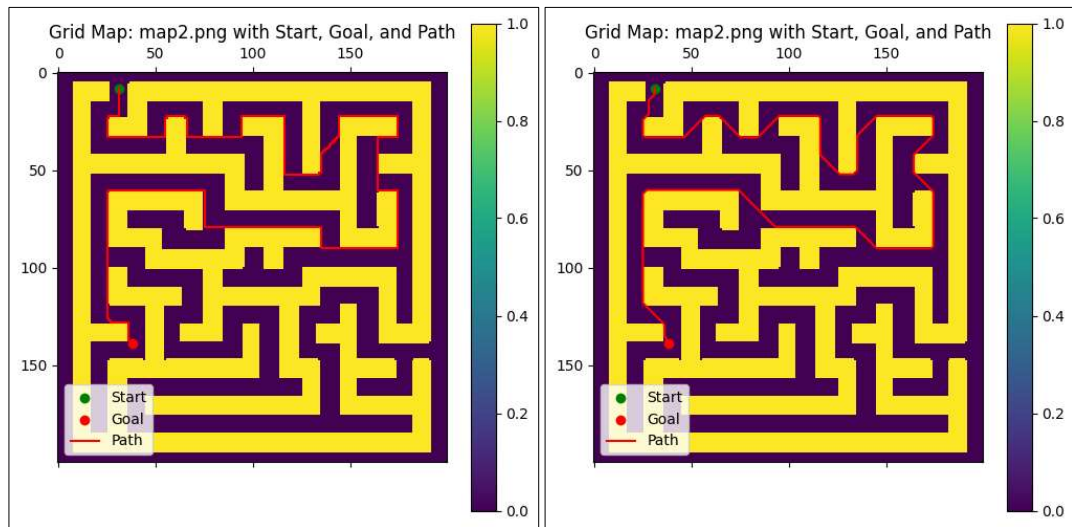
Inter a number between 0 and 3 to select environment and visibility: **1 (Enter)**

(Please refer to Appendix 3.2.2 map_1.png for output values of 4 point and 8 point connectivity with cost value)



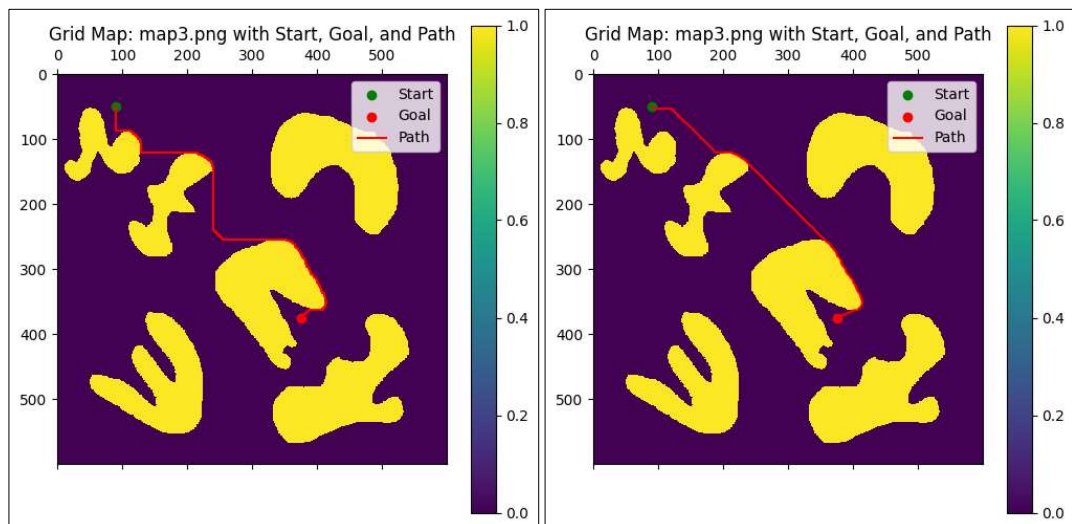
3.2.3 map_2.png

Enter a number between 0 and 3 to select environment and visibility: **2 (Enter)**
(Please refer to Appendix 3.2.3 map_2.png for output values of 4 point and 8 point connectivity with cost value)



3.2.4 map_3.png

Enter a number between 0 and 3 to select environment and visibility: **3 (Enter)**
(Please refer to Appendix 3.2.4 map_3.png for output values of 4 point and 8 point connectivity with cost value)



4.Problems Encountered

Problem 1 working with raw csv data

- **Challenge:** The data in the CSV files was unorganized, requiring identification and separation of start points, obstacles, goals, and visibility data.

- **Solution:**
 - a. Created dictionaries to structure the datasets.
 - b. Rearranged the data into well-defined categories such as start, goal, obstacles, and visibility.
 - c. Used this organized data to generate paths.

Problem 2: Using Visibility Data to Draw Paths

- **Challenge:** Incorporating visibility data to effectively compute and visualize paths in the graph was complex.
- **Solution:**
 - a. Parsed the visibility data to define connections between nodes.
 - b. Integrated this visibility information into the pathfinding logic of the A* algorithm.

Problem 3: Handling Multiple Maps and Environments

- **Challenge:** Generating outputs for all maps and environments required a flexible, user-friendly interface.
- **Solution:**
 - a. Developed a user-defined input interface within the main function.
 - b. Allowed users to seamlessly select environments and maps, ensuring smooth execution and output generation for all scenarios.

5. Conclusion

The lab exercises effectively demonstrated the use of the A* algorithm for finding the shortest path in various graph-based and grid-based scenarios. A* reliably identifies the shortest path by balancing exploration ($g(n)$) and goal-directed behavior ($h(n)$), to different environments, including datasets from .csv files and grid maps. Datasets were successfully loaded, parsed, and visualized to create graph structures and demonstrate visibility relationship. The implementation of 4-point and 8-point connectivity further highlighted how connectivity influences path selection, with 8-point connectivity offering more direct routes at the cost of increased computational complexity. The workflows, including user input handling, visualization, and computation, were organized for efficient processes. Overall, the lab highlighted the strengths of A* in solving diverse pathfinding challenges and provided foundational insights into graph traversal, heuristics, and connectivity.

Appendix

3.1 .csv files

3.1.1 env_0.csv and visibility_graph_env_0.csv

```

Inter a number between 0 and 3 to select environment and visibility: 0 (Enter)
Start Coordinates: (0.6, 8.0)
Target Coordinates: (10.3, 1.5)
Obstacles: {'1': [(3.0, 8.2), (4.0, 6.3), (2.0, 5.4)]}
Points: {0: (0.6, 8.0), 1: (3.0, 8.2), 2: (4.0, 6.3), 3: (2.0, 5.4), 4:
(10.3, 1.5)}

Visibility: [(0.0, 1.0), (0.0, 3.0), (1.0, 3.0), (1.0, 2.0), (1.0, 4.0), (2.0,
3.0), (2.0, 4.0), (3.0, 4.0)]

Shortest Path: [0, 3.0, 4.0]
    
```

3.1.2 env_1.csv and visibility_graph_env_1.csv

```
Inter a number between 0 and 3 to select environment and visibility: 1 (Enter)
Start Coordinates: (0.6, 8.0)
Target Coordinates: (5.3, 4.5)
Obstacles: {'3': [(3.5, 8.3), (2.2, 6.6), (3.3, 9.5), (5.5, 7.9)]}
Points: {0: (0.6, 8.0), 1: (3.5, 8.3), 2: (2.2, 6.6), 3: (3.3, 9.5), 4: (5.5, 7.9), 5: (5.3, 4.5)}
Visibility: [(0.0, 3.0), (0.0, 2.0), (1.0, 2.0), (1.0, 5.0), (1.0, 4.0), (2.0, 4.0), (2.0, 3.0), (2.0, 5.0), (3.0, 4.0), (4.0, 5.0)]
Shortest Path: [0, 2.0, 5.0]
```

3.1.3 env_2.csv and visibility_graph_env_2.csv

```
Inter a number between 0 and 3 to select environment and visibility: 2 (Enter)
Start Coordinates: (0.6, 8.0)
Target Coordinates: (16.3, 7.0)
Obstacles: {'1': [(3.0, 8.2), (4.0, 6.3), (2.0, 5.4)], '2': [(6.5, 8.3), (5.2, 6.6), (6.3, 4.7), (8.7, 6.3), (8.5, 7.9)], '3': [(13.5, 8.3), (12.2, 6.6), (13.3, 9.5), (15.5, 7.9)]}
Points: {0: (0.6, 8.0), 1: (3.0, 8.2), 2: (4.0, 6.3), 3: (2.0, 5.4), 4: (6.5, 8.3), 5: (5.2, 6.6), 6: (6.3, 4.7), 7: (8.7, 6.3), 8: (8.5, 7.9), 9: (13.5, 8.3), 10: (12.2, 6.6), 11: (13.3, 9.5), 12: (15.5, 7.9), 13: (16.3, 7.0)}
Visibility: [(0.0, 1.0), (0.0, 11.0), (0.0, 3.0), (1.0, 4.0), (1.0, 11.0), (1.0, 3.0), (1.0, 2.0), (1.0, 6.0), (1.0, 5.0), (2.0, 5.0), (2.0, 4.0), (2.0, 3.0), (2.0, 6.0), (3.0, 5.0), (3.0, 6.0), (4.0, 11.0), (4.0, 5.0), (4.0, 8.0), (5.0, 6.0), (6.0, 13.0), (6.0, 10.0), (6.0, 7.0), (7.0, 10.0), (7.0, 11.0), (7.0, 8.0), (8.0, 11.0), (8.0, 10.0), (9.0, 10.0), (9.0, 13.0), (9.0, 12.0), (10.0, 13.0), (10.0, 12.0), (10.0, 11.0), (11.0, 12.0), (12.0, 13.0)]
Shortest Path: [0, 1.0, 4.0, 8.0, 10.0, 13.0]
```

3.1.4 env_mx.csv and visibility_graph_env_mx.csv

```
Inter a number between 0 and 3 to select environment and visibility: 3 (Enter)
Start Coordinates: (14.0, 10.1)
Target Coordinates: (37.5, 23.8)
Obstacles: {'1': [(17.0, 16.5), (14.9, 16.9), (13.2, 18.4), (12.2, 21.5), (13.0, 25.0), (17.1, 25.4), (22.0, 24.4), (25.5, 25.0), (28.2, 24.3), (28.5, 21.4), (28.9, 18.5), (31.2, 16.8), (33.7, 16.3), (36.3, 15.9), (39.2, 16.5), (42.8, 17.6), (46.5, 16.8), (42.8, 15.2), (37.8, 12.9), (35.0, 12.0), (28.9, 12.3), (25.0, 12.8), (22.3, 15.0), (20.1, 17.7), (18.6, 20.6), (17.3, 22.5), (14.9, 23.2), (15.7, 21.5), (15.4, 19.6), (16.5, 18.0)]}
Points: {0: (14.0, 10.1), 1: (17.0, 16.5), 2: (14.9, 16.9), 3: (13.2, 18.4), 4: (12.2, 21.5), 5: (13.0, 25.0), 6: (17.1, 25.4), 7: (22.0, 24.4), 8: (25.5, 25.0), 9: (28.2, 24.3), 10: (28.5, 21.4), 11: (28.9, 18.5), 12: (31.2, 16.8), 13: (33.7, 16.3), 14: (36.3, 15.9), 15: (39.2, 16.5), 16: (42.8, 17.6), 17: (46.5, 16.8), 18: (42.8, 15.2), 19: (37.8, 12.9), 20: (35.0, 12.0), 21: (28.9, 12.3), 22: (25.0, 12.8), 23: (22.3, 15.0), 24: (20.1, 17.7), 25: (18.6, 20.6), 26: (17.3, 22.5),
```

```

27: (14.9, 23.2), 28: (15.7, 21.5), 29: (15.4, 19.6), 30: (16.5, 18.0), 31: (37.5,
23.8)}
Visibility: [(0.0, 20.0), (0.0, 21.0), (0.0, 22.0), (0.0, 23.0), (0.0, 24.0),
(0.0, 1.0), (0.0, 2.0), (0.0, 3.0), (0.0, 4.0), (1.0, 30.0), (1.0, 2.0), (1.0,
22.0), (2.0, 3.0), (2.0, 22.0), (3.0, 4.0), (4.0, 5.0), (5.0, 6.0), (6.0, 7.0),
(7.0, 8.0), (8.0, 9.0), (8.0, 31.0), (9.0, 10.0), (9.0, 17.0), (9.0, 31.0), (10.0,
31.0), (10.0, 11.0), (10.0, 17.0), (11.0, 31.0), (11.0, 9.0), (11.0, 12.0), (12.0,
31.0), (12.0, 9.0), (12.0, 13.0), (13.0, 31.0), (13.0, 9.0), (13.0, 14.0), (14.0,
15.0), (14.0, 31.0), (15.0, 16.0), (15.0, 31.0), (15.0, 9.0), (16.0, 31.0), (16.0,
17.0), (17.0, 31.0), (17.0, 18.0), (17.0, 19.0), (18.0, 19.0), (19.0, 20.0),
(20.0, 21.0), (21.0, 22.0), (22.0, 23.0), (23.0, 24.0), (23.0, 2.0), (24.0, 25.0),
(24.0, 1.0), (25.0, 26.0), (25.0, 1.0), (26.0, 27.0), (26.0, 1.0), (26.0, 24.0),
(27.0, 28.0), (27.0, 23.0), (28.0, 29.0), (28.0, 1.0), (28.0, 23.0), (29.0, 30.0),
(29.0, 22.0), (30.0, 22.0)]
Shortest Path: [0, 4.0, 5.0, 6.0, 7.0, 8.0, 31.0]

```

3.2 Grid Map

3.2.1 map_0.png

Inter a number between 0 and 3 to select environment and visibility: **0 (Enter)**

4-Point Connectivity:

```

Path: [(10, 10), (11, 10), (12, 10), (13, 10), (14, 10), (15, 10), (16, 10),
(17, 10), (18, 10), (19, 10), (20, 10), (21, 10), (22, 10), (23, 10), (24,
10), (25, 10), (26, 10), (27, 10), (28, 10), (29, 10), (30, 10), (30, 11),
(31, 11), (31, 12), (32, 12), (32, 13), (33, 13), (33, 14), (34, 14), (34,
15), (35, 15), (35, 16), (36, 16), (36, 17), (37, 17), (37, 18), (38, 18),
(38, 19), (39, 19), (40, 19), (40, 20), (41, 20), (41, 21), (42, 21), (43,
21), (43, 22), (44, 22), (44, 23), (45, 23), (45, 24), (46, 24), (46, 25),
(47, 25), (48, 25), (49, 25), (50, 25), (51, 25), (52, 25), (53, 25), (54,
25), (55, 25), (56, 25), (57, 25), (58, 25), (59, 25), (60, 25), (61, 25),
(62, 25), (63, 25), (64, 25), (65, 25), (66, 25), (67, 25), (68, 25), (69,
25), (70, 25), (71, 25), (72, 25), (73, 25), (74, 25), (75, 25), (76, 25),
(77, 25), (78, 25), (79, 25), (80, 25), (81, 25), (82, 25), (83, 25), (84,
25), (85, 25), (86, 25), (87, 25), (88, 25), (88, 26), (89, 26), (90, 26),
(91, 26), (91, 27), (92, 27), (92, 28), (93, 28), (93, 29), (94, 29), (94,
30), (94, 31), (95, 31), (95, 32), (95, 33), (96, 33), (96, 34), (96, 35),
(97, 35), (97, 36), (97, 37), (97, 38), (97, 39), (97, 40), (97, 41), (97,
42), (97, 43), (97, 44), (97, 45), (97, 46), (97, 47), (97, 48), (97, 49),
(97, 50), (97, 51), (97, 52), (97, 53), (97, 54), (97, 55), (97, 56), (97,
57), (97, 58), (97, 59), (97, 60), (97, 61), (97, 62), (97, 63), (96, 63),
(96, 64), (95, 64), (95, 65), (94, 65), (94, 66), (93, 66), (93, 67), (92,
67), (92, 68), (91, 68), (91, 69), (90, 69), (90, 70)]

```

Cost: 154.0

8-Point Connectivity:

```

Path: [(10, 10), (11, 11), (12, 12), (13, 13), (14, 14), (15, 15), (16, 16),
(17, 17), (17, 18), (17, 19), (17, 20), (17, 21), (17, 22), (17, 23), (17,
24), (17, 25), (17, 26), (17, 27), (17, 28), (17, 29), (17, 30), (17, 31),
(17, 32), (17, 33), (17, 34), (17, 35), (17, 36), (17, 37), (17, 38), (18,
39), (19, 40), (20, 41), (21, 42), (22, 43), (23, 44), (24, 45), (25, 46),

```

```
(25, 47), (26, 48), (27, 49), (28, 49), (29, 50), (30, 50), (31, 50), (32,
51), (33, 52), (34, 53), (35, 54), (36, 55), (37, 56), (38, 57), (39, 58),
(40, 59), (41, 59), (42, 59), (43, 60), (44, 61), (45, 62), (46, 63), (47,
64), (48, 65), (49, 66), (50, 67), (51, 68), (52, 69), (53, 70), (54, 71),
(55, 72), (56, 73), (57, 74), (58, 75), (59, 76), (60, 77), (61, 78), (62,
79), (63, 80), (64, 81), (65, 82), (66, 83), (67, 84), (68, 85), (69, 85),
(70, 86), (71, 86), (72, 87), (73, 87), (74, 87), (75, 87), (76, 87), (77,
86), (78, 85), (79, 84), (80, 83), (81, 82), (82, 81), (83, 80), (84, 79),
(85, 78), (86, 77), (87, 76), (88, 75), (88, 74), (89, 73), (89, 72), (89,
71), (90, 70)]
Cost: 133.5807358037434
```

3.2.2 map_1.png

```
Inter a number between 0 and 3 to select environment and visibility: 1 (Enter)
4-Point Connectivity:
Path: [(60, 60), (60, 59), (60, 58), (60, 57), (60, 56), (60, 55), (60, 54),
(60, 53), (60, 52), (60, 51), (60, 50), (60, 49), (60, 48), (60, 47), (60,
46), (60, 45), (60, 44), (60, 43), (60, 42), (59, 42), (58, 42), (57, 42),
(56, 42), (55, 42), (54, 42), (53, 42), (52, 42), (51, 42), (50, 42), (49,
42), (48, 42), (47, 42), (46, 42), (45, 42), (44, 42), (43, 42), (42, 42),
(42, 43), (41, 43), (41, 44), (41, 45), (41, 46), (41, 47), (41, 48), (41,
49), (41, 50), (41, 51), (41, 52), (41, 53), (41, 54), (41, 55), (41, 56),
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Cost: 240.0
8-Point Connectivity:
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Cost: 191.96551211459388

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3.2.3 map_2.png

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Inter a number between 0 and 3 to select environment and visibility: 2 (Enter)
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Cost: 632.0

8-Point Connectivity:

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Cost: 555.8477631085033
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3.2.4 map_3.png

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Inter a number between 0 and 3 to select environment and visibility: 3 (Enter)
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Cost: 688.0

8-Point Connectivity:

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Cost: 523.3940110268419
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